HMAC in an EtM configuration as described in section 6.3.1.

DEFINITION All the encryption algorithms allow the JWE header and IV to be included in the authentication tag without being encrypted. These are known as ***authenticated encryption with associated data (AEAD)*** algorithms.

The properties of the encryption methods are summarized in table 6.4. Larger key sizes offer more protection. The larger IV size of the CBC modes allows a much greater number of tokens to be encrypted before the key needs to change, but only extremely high-volume APIs are likely to reach the limit for GCM. The CBC modes also support larger authentication tags, although 128-bit tags are more than sufficient for short-lived tokens. The CBC modes need a larger key size because the key consists of two separate keys concatenated together: one for AES and one for HMAC.

TIP You should always plan to change encryption keys regularly even if they theoretically can be used for longer, to reduce the impact if a key is ever compromised.

Table 6.4 Comparison of JWE encryption methods

Encryption method	Key size (bits)	IV size (bits)	Usage limit	Tag size
`A128GCM`	128	96	4 billion	128
`A192GCM`	192	96	4 billion	128
`A256GCM`	256	96	4 billion	128